Case Study Report

Design and Creative Technologies

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Subject Code: ISY 503

Subject Name: Intelligent Systems

Assessment No.: 1

Title of Assessment: Case Study Report

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Date: Mar 2026

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Application of Intelligent System in Natural Language Processing (NLP)

1. Introduction

1.1. Case Study Overview

Digital health interventions (DHIs), including smartphone apps, online coaching platforms, and web-based therapy programs, generate large volumes of unstructured text through participant journals, coach-user messages, and self-monitoring logs. Extracting clinically meaningful insights from this text manually is time-intensive and does not scale to the thousands of users these platforms serve.

Funk et al. (2020) addressed this gap by proposing the Digital Health Intervention Text Analytics (DHITA) framework: a reusable two-step natural language processing (NLP) pipeline designed to automate feature extraction from DHI text and apply those features to supervised machine learning models for clinical outcome prediction. The framework was validated on data from two randomized controlled trials (RCTs) targeting eating disorders among college-age women.

1.2. Problem Significance

Mental health conditions affect over 970 million people globally, yet specialist care access remains severely constrained (World Health Organization, 2022). DHIs offer a scalable route to evidence-based support, but their clinical value depends on monitoring user

progress at scale - a task manual review cannot sustain. Intelligent NLP systems can bridge this gap by transforming unstructured language into actionable clinical signals.

1.3. Report Structure

This report examines the DHITA framework as a case study in applying intelligent systems to high-stakes health data. Section 2 provides technical background on the system's architecture, ML models, and feature engineering methods. Section 3 describes the research methodology, data sources, and ethical considerations. Section 4 presents quantitative results, Section 5 discusses limitations, and Section 6 offers recommendations for improvement.

2. Background

2.1. Intelligent Systems Overview

DHITA is a supervised ML system for text analytics operating as a two-step pipeline: (1) feature engineering (transforming raw DHI text into numerical representations) and (2) predictive modeling (applying those features to trained classifiers and regressors for clinical outcome prediction).

2.2. Machine Learning Models

Two predictive models are deployed. **Model A**: a Random Forest classifier (Breiman, 2001) with 200 decision trees that predicts session-level symptom severity from post-session text entries. Random Forest is well-suited to this task because it handles high-dimensional, mixed-type feature sets without requiring feature scaling and reduces overfitting through ensemble

aggregation. **Model B**: a Logistic Regression classifier that predicts long-term clinical outcomes (eating disorder remission) at six-month follow-up. Logistic Regression was chosen for interpretability, which is critical in clinical contexts where prediction rationale must be auditable.

Feature selection is performed using the Least Absolute Shrinkage and Selection Operator (LASSO) with 50-fold cross-validation (Tibshirani, 1996). LASSO imposes L1 regularization that drives irrelevant feature coefficients to zero, selecting a compact interpretable subset from the 241 available features, necessary given the limited sample size.

2.3. Methods and Techniques

The feature engineering layer combines seven representation strategies to produce 241 features per user. (convert to a table)

- *Metadata features* (n=5) capture session-level statistics such as message length and response time.
- *Word frequency features* (n=79) apply minimum/maximum occurrence thresholds to filter vocabulary by clinical relevance.
- *GloVe word embeddings* (n=50) average 50-dimensional pre-trained vectors across tokens to produce a fixed-length document representation (Pennington et al., 2014).
- *Part-of-speech (POS) tagging* (n=44) uses Apache OpenNLP to count grammatical category frequencies at approximately 95% accuracy.
- *Latent Dirichlet Allocation (LDA) topic modeling* (n=10) extracts eight latent themes from the corpus (Blei et al., 2003).

- *Sentiment lexicons* (n=30) leverage NRC Emotion Lexicon, AFINN, and Bing to quantify affective tone.
- Area Under the ROC Curve (AUC) serves as the primary evaluation metric, selected for its robustness to class imbalance.

2.4. Related Work

Prior health NLP work focused on clinical notes and social media (Sarker, 2021); DHIs present a distinct type: longitudinal coach-participant dialogues within structured therapeutic contexts. Earlier DHI studies used isolated sentiment scores or word counts. DHITA is the first framework to systematically combine all major NLP feature classes in a single replicable pipeline.

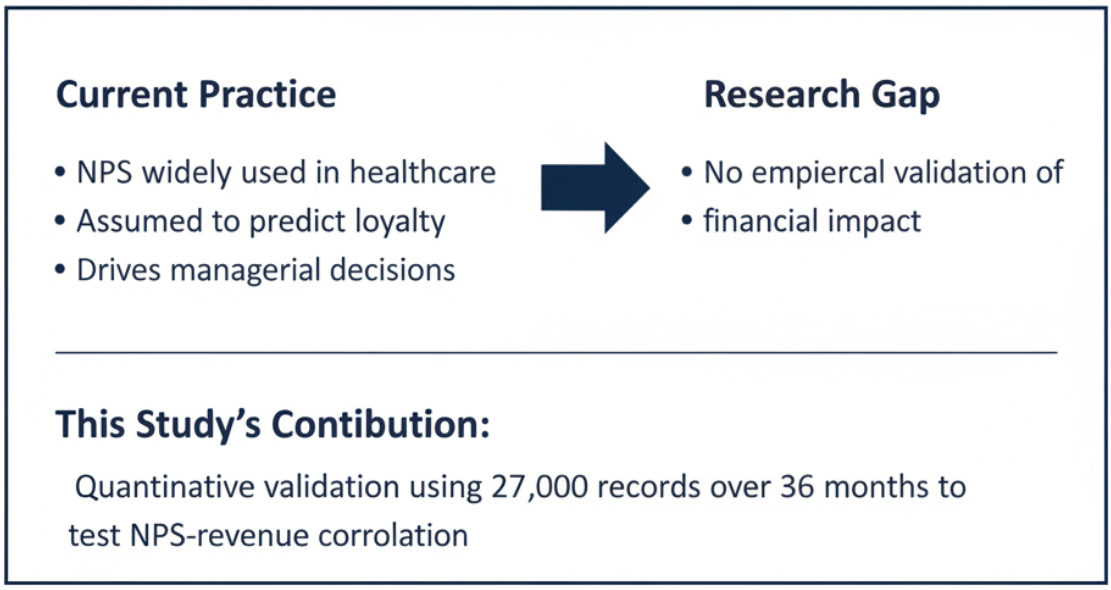

Figure 1 – Lorem ipsum.

3. Method

3.1. Research Methodology

The study employed a retrospective observational design, re-analyzing text data from two previously conducted RCTs: the Self-Monitoring of Body, Eating, and Dieting (SBED) program and the Healthy Body Image (HBI) program, both targeting eating disorders in college-age women. This cross-sectional approach allowed evaluation of DHITA's predictive validity against established clinical measures without new data collection.

3.2. Data Sources

The dataset comprised 37,228 intervention text snippets from SBED participant journals and 4,285 coach-user messages from the HBI program, contributed by 372 users in total. Of these, 100 users with complete six-month follow-up assessments using the Eating Disorder Examination Questionnaire (EDE-Q) were used for outcome prediction. The study population was homogeneous: predominantly college-age females with subclinical eating disorders — a deliberate clinical design choice that simultaneously limits external validity.

Text preprocessing followed standard NLP practices: tokenization, stemming (Porter stemmer), and vocabulary filtering via MINOCC/MAXOCC thresholds to remove rare and overly common tokens. GloVe vectors were averaged across tokens per document. LDA was trained on the full corpus to extract eight latent topics. Model evaluation used a within-user protocol for Model A (training and testing on the same user's temporal data) and a cross-user

protocol for Model B (evaluation across 100 users). LASSO with 50-fold cross-validation was applied for feature selection prior to final model fitting.

3.3. Ethical Considerations

The dataset contains sensitive mental health disclosures — eating disorder diaries and private coach messages. Privacy is maintained at the aggregate feature level, but re-identification risk persists in small, homogeneous cohorts; participants consented to RCT participation, not secondary NLP analysis. Training exclusively on college-age females introduces demographic bias, limiting equitable performance across other populations. LASSO provides partial interpretability, but Random Forest lacks native explainability, requiring post-hoc tools for clinical audit. Most critically, AUC 0.57–0.72 is insufficient for autonomous clinical decisions; deploying predictions without human-in-the-loop oversight risks triage errors and harm to vulnerable users.

3.4. Implementation Process

The pipeline was implemented using standard scientific Python tooling, with Apache OpenNLP providing POS tagging. No deep learning frameworks were required; computational demands were modest and hardware constraints are not reported as a limiting factor.

4. Results

4.1. Performance Metrics

Model A (symptom severity prediction, Random Forest) achieved an AUC of 0.72 under within-user evaluation — where the model trains and tests on the same individual's temporal data. Under the more rigorous cross-user protocol, AUC dropped to 0.57, marginally above the random baseline of 0.50. Model B (six-month outcome prediction, Logistic Regression) identified approximately 10 key predictive features following LASSO selection. Both models outperformed the random baseline; within-user AUC of 0.72 represents the study's strongest result.

4.2. Key Findings

The most discriminative features for outcome prediction centered on thematic and behavioral signals: body-image and diet-related language, references to program content ("help," "program" terms), and temporal engagement metrics such as message response rate and response time. This suggests that linguistic engagement with program content — not just emotional tone — predicts treatment trajectory.

The full feature set comprised 241 dimensions: metadata (5), communication statistics (23), word frequency (79), GloVe embeddings (50), POS tags (44), LDA topics (10), and sentiment scores (30). Correlation analysis of the feature matrix showed low inter-feature correlation, confirming that the seven feature groups contribute non-redundant information.

4.3. Visual Analysis

The original paper reports ROC curves for both models, with Model A's within-user curve showing the most pronounced departure from the diagonal. Feature importance rankings derived from LASSO coefficients are documented in the paper, with the body/diet/program lexical cluster and response time metrics appearing as the highest-weighted predictors.

5. Discussion

5.1. Relevance and Significante

DHITA demonstrates that NLP-derived features from digital health text carry predictive signal for clinical outcomes. The framework's modular design is a genuine contribution: by standardizing the feature engineering and modeling pipeline, it provides a replicable scaffold for DHI researchers, reducing the barrier to applying NLP in new health domains. The proof-of-concept AUC of 0.72 for symptom severity prediction validates the core hypothesis that linguistic signals in DHI text are clinically informative.

5.2. Technical Obstacles

Technical Obstacles: The within-user/cross-user performance gap — from AUC 0.72 to 0.57 — is the study's most critical finding. It reveals that the model learns idiosyncratic patterns specific to individuals rather than generalizable population-level signals, fundamentally limiting scalability to new users.

Methodological Limitations: The dataset is drawn exclusively from college-age females with eating disorders enrolled in structured RCTs. This narrow population restricts external validity and raises equity concerns: the model may perform poorly on demographically distinct groups. The 100-user outcome prediction sample is small, increasing variance in cross-user evaluation.

Constraints Reported in Article: The authors explicitly acknowledge the preliminary status of the results, the absence of deep learning methods, and the need for validation across additional conditions and populations.

5.3. Comparison with State-of-the-Art

The study predates the broad adoption of transformer-based NLP. Since 2020, BERT and its clinical variants (e.g., BioBERT, MentalBERT) have substantially advanced clinical text mining. Averaged GloVe embeddings cannot capture contextual meaning; replacing them with transformer representations would likely yield measurable AUC improvements and is the most actionable technical upgrade available today.

6. Recommendations

6.1. Technical Improvements

Replacing GloVe embeddings with contextual representations from transformer models — such as MentalBERT or BioBERT — would capture semantic nuance that averaged static vectors cannot represent, likely pushing AUC toward the ≥0.85 threshold required for clinical

decision support. Expanding the corpus to diverse populations (age groups, genders, comorbid conditions) would directly address the generalizability gap.

6.2. Ethical and Practical Recommendations

Any clinical deployment must incorporate mandatory human-in-the-loop review; no triage decision should rely solely on algorithmic output. SHAP or LIME explainability tools would enable clinicians to audit predictions. Consent frameworks must be updated to explicitly cover secondary NLP analysis. Integrating passive sensing signals — app usage frequency, session timing, response latency — alongside text features could improve predictive power without requiring additional sensitive disclosures. Future work could explore federated learning to enable multi-site training while preserving participant privacy.

7. Conclusion

Funk et al.'s DHITA framework represents a meaningful step toward scalable clinical insight extraction from digital health text. By combining seven NLP feature engineering strategies with interpretable ML models and LASSO feature selection, the framework achieves an AUC of 0.72 for within-user symptom severity prediction — a promising proof-of-concept for NLP in mental health. However, the cross-user performance drop to AUC 0.57, the narrow study population, and the absence of contextual deep learning methods limit immediate clinical translation. The field has since advanced considerably; replicating DHITA with transformer-based embeddings and diverse, multicondition datasets represents the most productive path forward. Any deployment must prioritize human oversight to ensure predictive models complement rather than replace clinical judgment.

8. Appendices

8.1. Appendix A – X

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End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (GPT-5)
- Anthropic Claude Sonnet 4.6

Both have been used to assist with outlining, refining structure, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1.

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I confirm that the use of the AI tool has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

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